



Handling Tables

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(expleo)

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Web Tables



Introduction

- Web table in Selenium is a WebElement just like any other popular WebElements like text boxes, radio buttons, checkboxes, drop-down menus, etc.
- Web table and its contents can be accessed by using the WebElement functions along with Selenium locators to identify the element (row/column) on which the operation needs to be performed.
- A table consists of rows and columns. Below are some of the important tags associated with a web table:
 - < table > – Defines an HTML table
 - < th > – Contains header information in a table
 - < tr > – Defines a row in a table
 - < td > – Defines a column in a table

Web Tables

Example – Table Creation

```
<html>
  <table border="5">
    <thead>
      <tr>
        <th>Name</th>
        <th>Age</th>
        <th>Place</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td>Arun</td>
        <td>35</td>
        <td>Coimbatore</td>
      </tr>
      <tr>
        <td>Vijay</td>
        <td>37</td>
        <td>Chennai</td>
      </tr>
    </tbody>
  </table>
</html>
```



A screenshot of a web browser window showing the rendered HTML table. The browser's address bar displays the file path: C:/Users/Arun%20Kumar/Downloads/table.html. The table has a thick black border and contains the following data:

Name	Age	Place
Arun	35	Coimbatore
Vijay	37	Chennai

Types of Web Tables



Types of Web Tables

Web Table Types

- There are two broad categories of tables namely:

1. Static Web Table

- As the name indicates, the information in the table is static in nature.
- These tables have fixed data that remains unchanged throughout.

2. Dynamic Web Table

- The information displayed in the table is dynamic.
- These tables have data that changes over time, and hence the number of rows and columns might also change depending upon the data shifts.
- E.g. Detailed Product information on e-commerce websites, sales reports, etc.

Handling Web Tables in Selenium



Inspecting table and its attributes using XPath

- It is very seldom for a web designer to provide an id or name attribute to a certain cell in the table.
- Therefore, we cannot use the usual methods such as "By.id()", "By.name()", or "By.cssSelector()".
- In this case, the most reliable option is to access them using the "By.xpath()" method.
- Refer the <http://omayo.blogspot.com/> for the testing purpose.

Handling Web Tables in Selenium

Inspecting table and its attributes using XPath

- Xpath is `//table[@id='table1']` - to locate complete table
- Xpath is `//table[@id='table1']/thead` - to locate table heading section
- Xpath is `//table[@id='table1']/tbody` - to locate table body section

- Xpath is `//table[@id='table1']/thead/tr/th`
or
- Xpath is `//table[@id='table1']//th` - to locate all table headings

- Xpath is `//table[@id='table1']/tbody/tr/td`
or
- Xpath is `//table[@id='table1']//td` - to locate all the table data

Handling Web Tables in Selenium

Inspecting table and its attributes using XPath

- Xpath is `//table[@id='table1']/tbody/tr[1]` - to locate first row in the table body
- Xpath is `//table[@id='table1']/tbody/tr[2]` - to locate second row in the table body
- Xpath is `//table[@id='table1']/tbody/tr[3]` - to locate third row in the table body

- Xpath is `//table[@id='table1']/tbody/tr[1]/td` - to locate all the data in the first row
- or
- Xpath is `//table[@id='table1']//tr[1]//td` - to locate all the data in the first row

- Xpath is `//table[@id='table1']//tr[2]//td` - to locate all the data in the second row

Handling Web Tables in Selenium

Inspecting table and its attributes using XPath

- Xpath is `//table[@id='table1']/tbody/tr[1]/td[1]` - to locate the data in 1st row and 1st column
or
- Xpath is `//table[@id='table1']//tr[1]//td[1]` - to locate the data in 1st row and 1st column
- Xpath is `//table[@id='table1']//tr[2]//td[2]` - to locate the data in 2nd row and 2nd column
- Xpath is `//table[@id='table1']//tr[3]//td[2]` - to locate the data in 3rd row and 2nd column
- Xpath is `//table[@id='table1']//tr//td[3]` - to locate all the data in the 3rd column

Handling Web Tables in Selenium

Handling Static Table – Printing Table Headings

```
import time

from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://omayo.blogspot.com/")

# To print table headings

table_headings = driver.find_elements(By.XPATH, value="//table[@id='table1']//th")

for headings in table_headings:
    print(headings.text)

time.sleep(5)
```

Output

Output:

PROBLEMS

OUTPUT

DEBUG CONSOLE

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[Done] exited with code=1 in 4.475 seconds

[Running] python -u "c:\Users\Arun Kumar\VSCodePython\table.py"

[Done] exited with code=0 in 11.871 seconds

[Running] python -u "c:\Users\Arun Kumar\VSCodePython\table.py"

Name

Age

Place

[Done] exited with code=0 in 11.926 seconds

Handling Static Table – Printing all Table data

```
import time

from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://omayo.blogspot.com/")

#To retrieve complete table data

table_data = driver.find_elements(By.XPATH, value="//table[@id='table1']//td")

for data in table_data:
    print(data.text)
```


Handling Web Tables in Selenium

Output

Output:

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Code



[Done] exited with code=1 in 0.142 seconds

[Running] python -u "c:\Users\Arun Kumar\VSCodePython\table.py"

Kishore

22

Delhi

Manish

25

Pune

Praveen

29

Bangalore

Dheepthi

31

Mumbai

[Done] exited with code=0 in 13.12 seconds

Inspecting table and its attributes using XPath

- It is very seldom for a web designer to provide an id or name attribute to a certain cell in the table.
- Therefore, we cannot use the usual methods such as "By.id()", "By.name()", or "By.cssSelector()".
- In this case, the most reliable option is to access them using the "By.xpath()" method.

Handling Web Tables in Selenium

Handling Static Table – To retrieve first row from table

```
import time

from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://omayo.blogspot.com/")

#To retrieve first row from table

table_row1data = driver.find_elements(By.XPATH, value="//table[@id='table1']/tbody/tr[1]/td")

for data in table_row1data:
    print(data.text)
```

Handling Web Tables in Selenium

Output

Output:

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Code



```
[Running] python -u "c:\Users\Arun Kumar\VSCodePython\table.py"
```

```
Kishore
```

```
22
```

```
Delhi
```

```
[Done] exited with code=0 in 11.846 seconds
```

Handling Web Tables in Selenium

Handling Static Table – To retrieve second row third column data from table

```
import time

from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://omayo.blogspot.com/")

#To retrieve second row third column data from table

table_row2column3data = driver.find_element(By.XPATH, value="//table[@id='table1']//tr[2]/td[3]")

print(table_row2column3data.text)
```

Handling Web Tables in Selenium

Output

Output:

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

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Code



```
[Running] python -u "c:\Users\Arun Kumar\VSCodePython\table.py"
```

```
Pune
```

```
[Done] exited with code=0 in 12.771 seconds
```

Handling Web Tables in Selenium

Handling Static Table – To retrieve all data in second column

```
import time

from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://omayo.blogspot.com/")

#To retrieve all data in second column

table_column2data = driver.find_elements(By.XPATH, value="//table[@id='table1']/tbody/tr/td[2]")

for data in table_column2data:
    print(data.text)
```

Handling Web Tables in Selenium

Output

Output:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Code ▼

```
[Running] python -u "c:\Users\Arun Kumar\VSCodePython\table.py"  
22  
25  
29  
31  
  
[Done] exited with code=0 in 11.93 seconds
```


Handling Web Tables in Selenium

Handling Static Table – To find number of rows and columns in the table

```
import time

from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://omayo.blogspot.com/")

#To find number of rows and columns in the table

rows = driver.find_elements(By.XPATH, value="//table[@id='table1']//tr")
print(len(rows))

columns = driver.find_elements(By.XPATH, value="//table[@id='table1']//th")
print(len(columns))
```

Output

Output:

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Code

```
[Running] python -u "c:\Users\Arun Kumar\VSCodePython\table.py"
```

```
5
```

```
3
```

```
[Done] exited with code=0 in 13.252 seconds
```

Handling Web Tables in Selenium

Handling Static Table – To retrieve entire table

```
import time

from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://omayo.blogspot.com/")

#To retrieve entire table

rows = driver.find_elements(By.XPATH, value="//table[@id='table1']//tr")
rows_count = len(rows)

columns = driver.find_elements(By.XPATH, value="//table[@id='table1']//th")
columns_count = len(columns)
```

Handling Web Tables in Selenium

Handling Static Table – To find number of rows and columns in the table

```
for r in range(1,rows_count+1):
    for c in range(1,columns_count+1):
        if r == 1:
            data = driver.find_element(By.XPATH,
value="//table[@id='table1']//tr["+str(r)+"]//th["+str(c)+"]")
            print(data.text,end=' ')
        else:
            data = driver.find_element(By.XPATH, value="//table[@id='table1']//tr["+str(r-
1)+"]//td["+str(c)+"]")
            print(data.text,end=' ')
    print()
```

Handling Web Tables in Selenium

Output

Output:

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Code



```
[Running] python -u "c:\Users\Arun Kumar\VSCodePython\table.py"
```

```
5
```

```
3
```

```
Name Age Place
```

```
Kishore 22 Delhi
```

```
Manish 25 Pune
```

```
Praveen 29 Bangalore
```

```
Dheepthi 31 Mumbai
```

```
[Done] exited with code=0 in 13.51 seconds
```

Handling Web Tables in Selenium

Handling Dynamic Table – problem in accessing Changing data

```
import time

from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://thinking-tester-contact-list.herokuapp.com/")

driver.find_element(By.ID, value="email").send_keys("aa.83@gmail.com")
driver.find_element(By.ID, value="password").send_keys("Test143$")
driver.find_element(By.ID, value="submit").click()

#problem in accessing data in dynamic table

required_data = driver.find_element(By.XPATH, value="//table[@id='myTable']/tr[2]")
print(required_data.text)
```

Handling Web Tables in Selenium

Output

Output:

PROBLEMS

OUTPUT

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Code

```
[Running] python -u "c:\Users\Arun Kumar\VSCodePython\dynamicTable2.py"
```

```
Babu A 1978-09-12 babu@gmail.com 9876549871 12 west street pune Pune Mumbai 678213 India
```

```
[Done] exited with code=0 in 18.676 seconds
```

```
[Running] python -u "c:\Users\Arun Kumar\VSCodePython\dynamicTable2.py"
```

```
Ajay A 1982-03-08 ajay08@gmail.com 8976512340 Plot 1 Ram Nagar Bangalore Karnataka 691010 India
```

Handling Dynamic Table – Retrieving row data

```
import time

from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://thinking-tester-contact-list.herokuapp.com/")

driver.find_element(By.ID, value="email").send_keys("aa.83@gmail.com")
driver.find_element(By.ID, value="password").send_keys("Test143$")
driver.find_element(By.ID, value="submit").click()
```


Handling Dynamic Table – Retrieving row data

```
#Retrieving row data from dynamic table

expected_name = "Babu A"
contact_names = driver.find_elements(By.XPATH, value="//table[@id='myTable']/tr/td[2]")
contact_count = len(contact_names)
print(contact_count)

for name in contact_names:
    print(name.text)

i = 1

for name in contact_names:
    if name.text .__eq__(expected_name):
        actual_rowdata = driver.find_elements(By.XPATH, value="//table[@id='myTable']/tr["+str(i)+"]")
        for actname in actual_rowdata:
            print(actname.text)
        i=i+1
```

Handling Web Tables in Selenium

Output

Output:

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Code



[Running] python -u "c:\Users\Arun Kumar\VSCodePython\dynamicTable2.py"

4

A A

Ajay A

Babu A

raj k

Babu A 1978-09-12 babu@gmail.com 9876549871 12 west street pune Pune Mumbai 678213 India

[Done] exited with code=0 in 16.395 seconds